

Decent Coaching Classes

VIII SB

Mind Your Math - II

Test: Chapters 4, 5, 6 and 7.1

Time: 1 Hour

Marks: 30

Note:

- (1) All questions are compulsory. The number to the right of each question indicates full marks.
- (2) Write proper alternative letter for questions in Q.1(A).
- (3) Draw neat figures wherever necessary.

Q.1 (A) Fill in the Blanks.

[4]

- (1) The centroid of a triangle divides each median in the ratio _____ from the vertex.
- (2) Which of the following is the correct expansion of $(a - b)^3$?
- (3) If $x^2 + 8x + 15 = (x + a)(x + b)$, then the values of a and b are _____.
- (4) If x varies directly as y, and $x = 8$ when $y = 2$, the constant of variation k is _____.

Q.1 (B) Solve the following. (Attempt any FOUR)

[4]

- (1) State the formula for $a^3 - b^3$.
- (2) Write $(p + q)^3$ using the expansion formula.
- (3) In $\triangle XYZ$, seg $XR \perp$ seg YZ . State what seg XR is called.
- (4) Factorise: $x^2 + 7x + 12$
- (5) If y varies directly as x, write the equation of variation using constant k.

Q.2 Solve. (Attempt any TWO)

[4]

- (1) $(2x + 3y + 4)^2$.
- (2) Factorise $5x^2 + 16x + 3$.
- (3) x varies directly as y. Complete the following table.

x	4	6		10
y	20		40	

Q.3 Solve the following. (Attempt any FOUR)

[8]

- (1) Expand: $(2p - 5)(2p + 7)$
- (2) In $\triangle ABC$, medians AD , BE and CF intersect at the centroid G . If $AG = 10$ cm, find $l(GD)$.
- (3) Factorise: $3x^2 - x - 4$
- (4) The cost of 15 pens is ₹225. If the cost varies directly as the number of pens, find the cost of 36 pens.
- (5) Simplify using formula: $(x + 3)^3 + (x - 3)^3$
- (6) Factorise: $8a^3 - 125b^3$

Q.4 Solve the following. (Attempt any TWO)

[10]

- (1) In $\triangle ABC$, medians AD , BE and CF intersect at the centroid G .
 - (i) If $AD = 12$ cm, find $l(AG)$ and $l(GD)$.
 - (ii) If $l(BG) = 8$ cm, find $l(GE)$ and $l(BE)$.
 - (iii) State the property of the centroid of a triangle used above.
- (2) (i) Factorise: $2x^2 - 7x - 15$
(ii) Factorise: $27p^3 + 1$

(iii) Simplify: $(a^2 + 6a + 8)/(a^2 + 9a + 20) \div (a^2 + 2a - 8)/(a^2 + 3a - 10)$

(3) The total remuneration paid to labourers varies directly with the number of labourers. If the remuneration of 6 labourers is ₹1500,

(i) Find the constant of variation and the equation of variation.

(ii) Find the remuneration of 25 labourers.

(iii) Find the number of labourers if the remuneration paid is ₹4250.

(4) (i) Expand: $(3x + 2y + 5)^2$

(ii) Using the formula $(a + b)^3 + (a - b)^3 = 2a^3 + 6ab^2$, simplify $(2m + 3n)^3 + (2m - 3n)^3$

——— **Best of Luck** ———